

p values

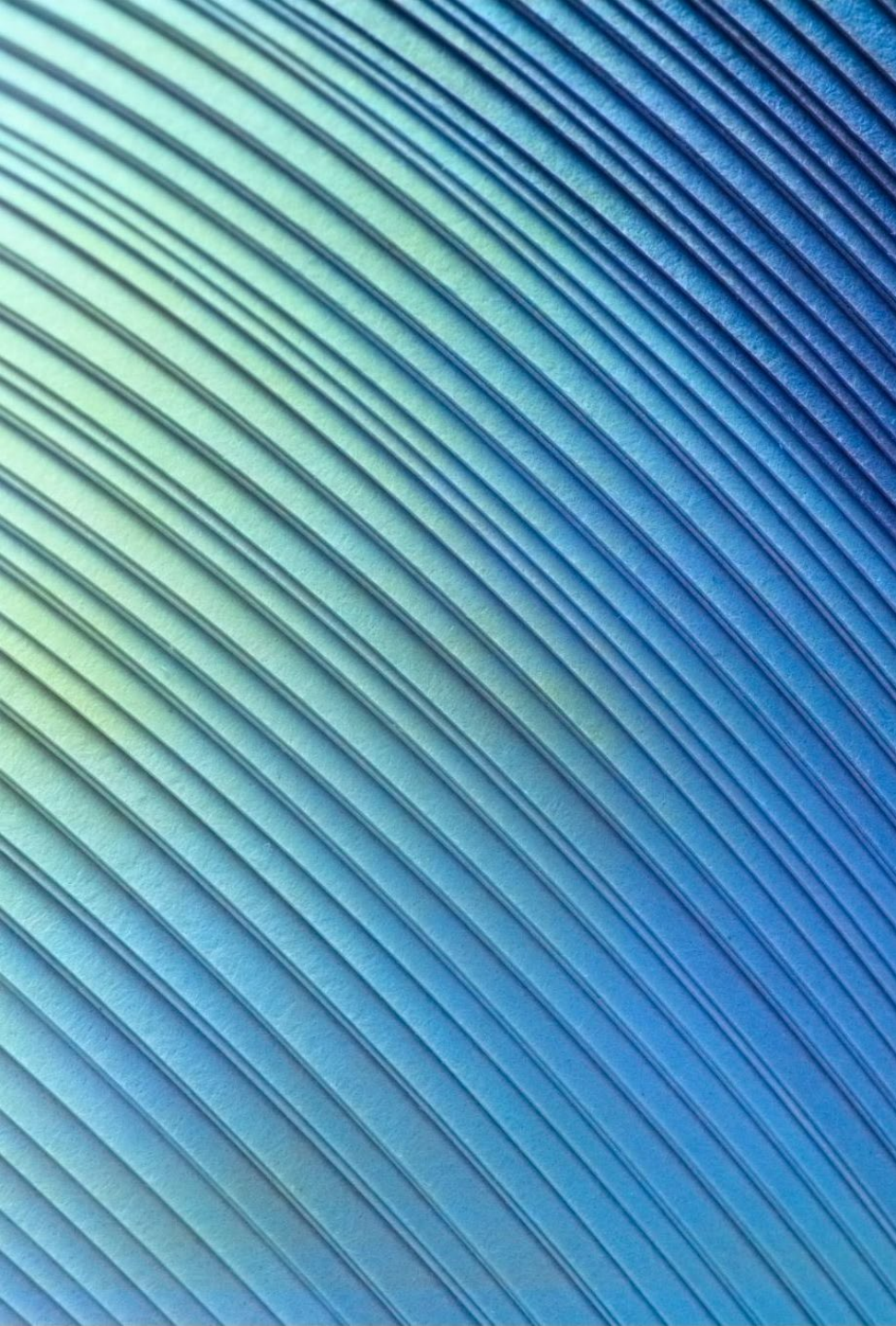


p values

p value is a concept widely used (and abused) in statistics.

We'll learn what a p value is, what it isn't, and how it is employed in standard hypothesis testing situations.

We'll discover how to compute p values from several distributions.



Quantifying the Unusual

In standard hypothesis testing, we employ a strategy that is based on falsifying the opposite of what we are trying to show.

For example, suppose we believe that giving a drug to a group of people will make them unusual.

We employ the following logic.

Quantifying the Unusual

- We begin by stating a hypothesis that is the opposite of what we believe.
- That is, we state as our "statistical null hypothesis" that the group is not unusual.

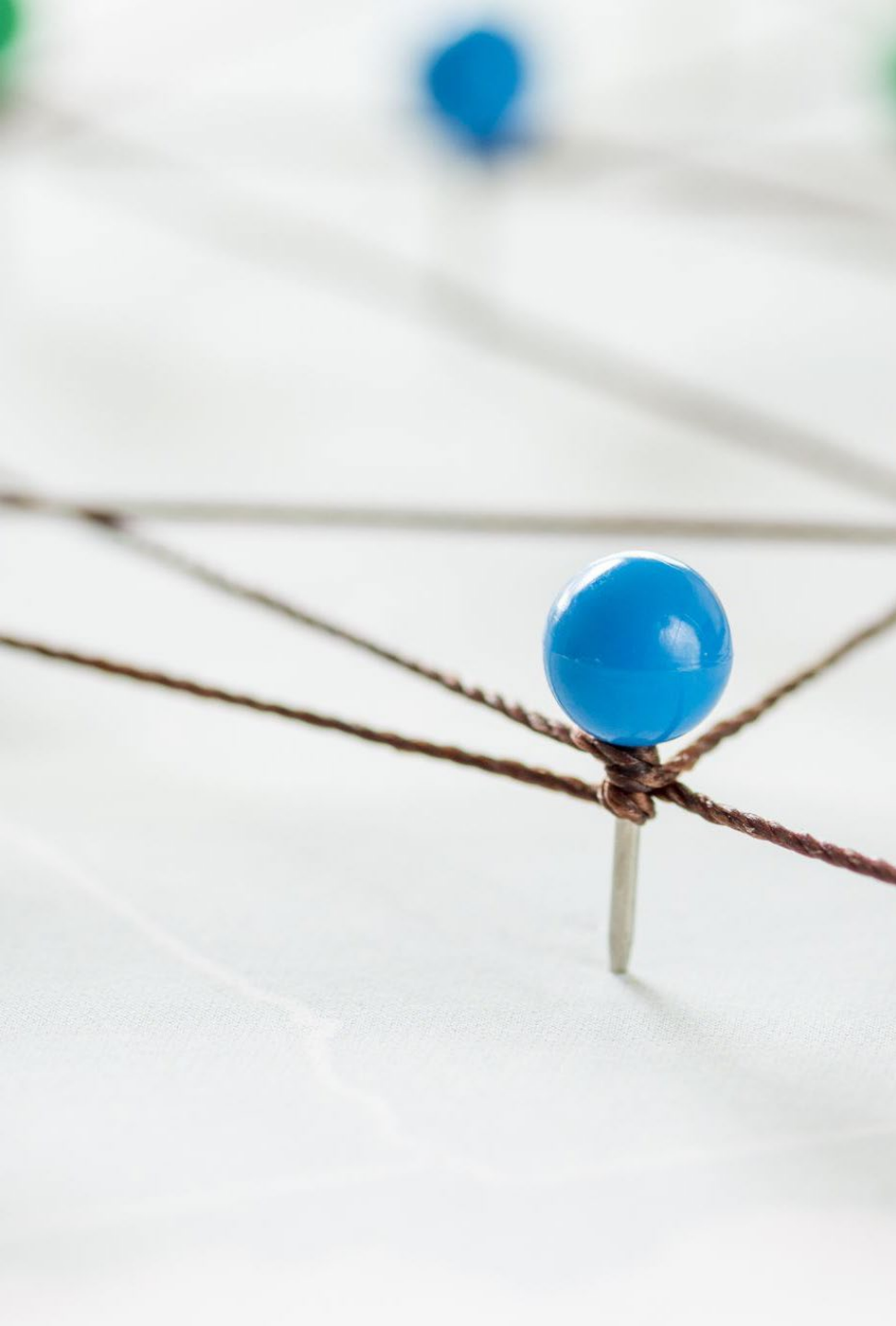


Quantifying the Unusual

What we then seek to do is show that, in fact, the group is unusual,

- thereby falsifying the opposite of what we wish to show, and
- demonstrating the truth of what we wish to show.

So, we need a way of demonstrating that an observation (our data) is unusual enough to be unlikely to have occurred by chance.



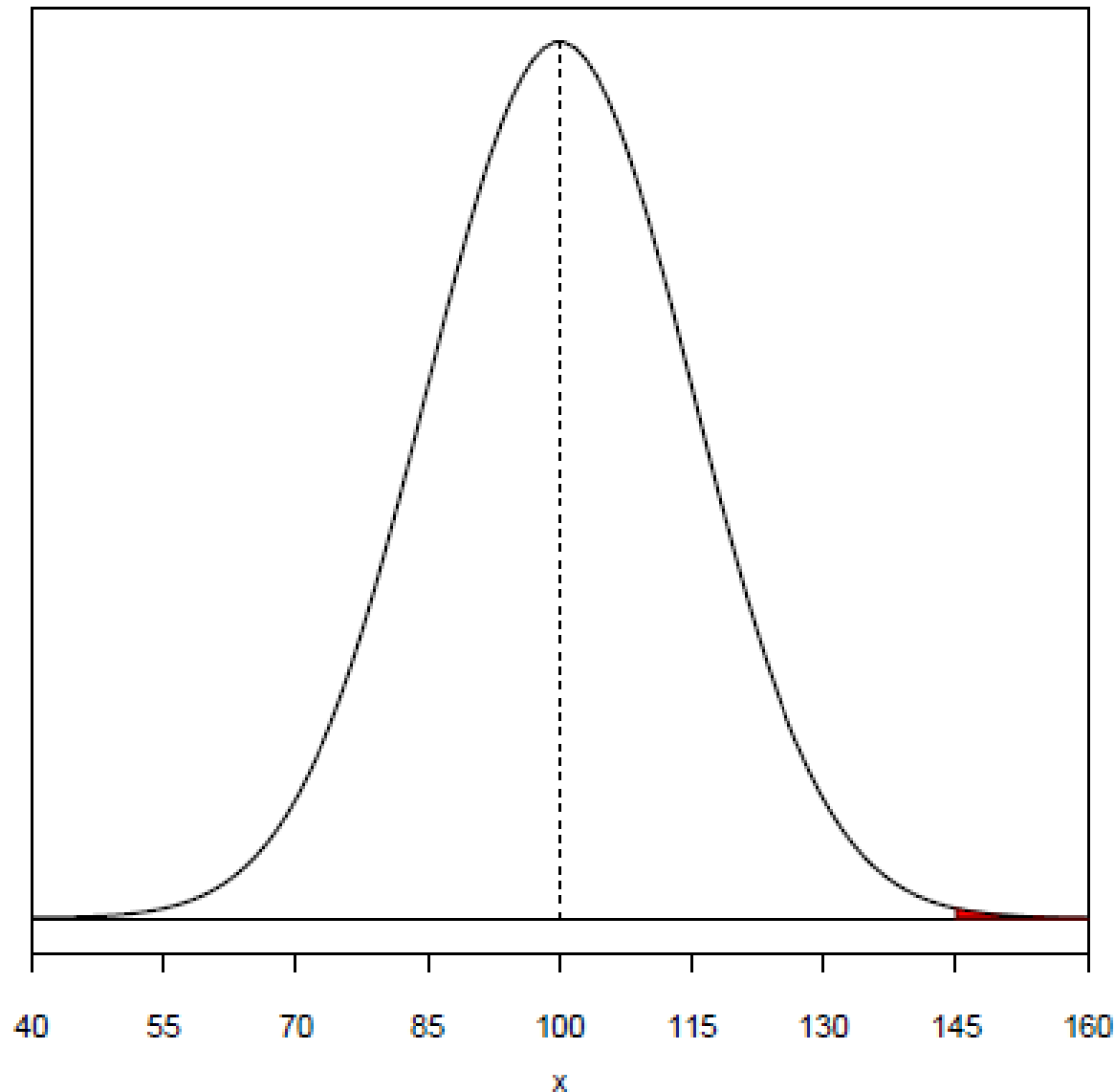
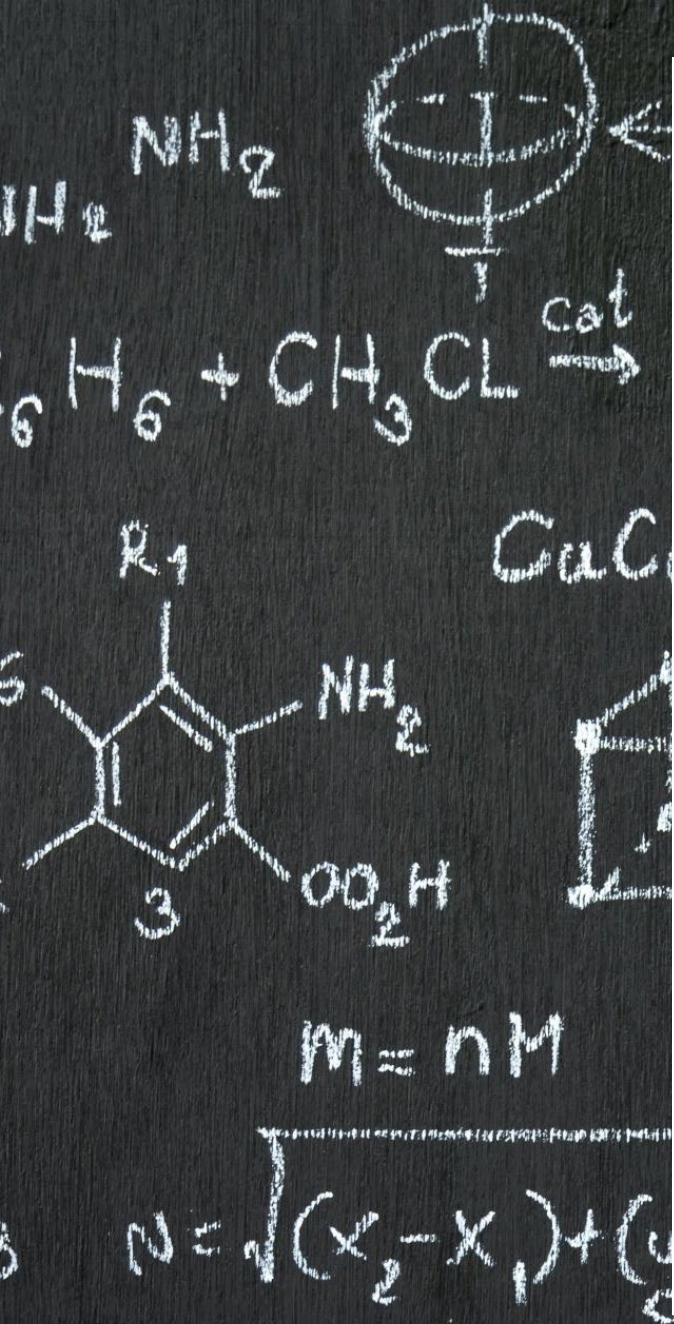
Quantifying the Unusual

- Suppose a population of scores has a probability distribution.
- How can we decide if an observation is unusual, or extreme?
- For example, suppose IQ scores are normally distributed with a mean of
- 100 and a standard deviation of 15. Is an IQ of 145 unusual, or extreme?
- How would you decide?

Quantifying the Unusual

One way of quantifying how unusual a result is involves a probability calculation.

A result X is unusual if the probability of a result more extreme than X is small.



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How Extreme? Two-Sided vs. One-Sided

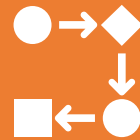
There is a catch. Suppose you consider an IQ of 100 to be normal.

Wouldn't it be correct to say that an IQ of 55 is just as unusual as an IQ of 145?

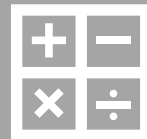
The probability of being as extreme as or more extreme than 145 in the positive direction is exactly the same as

- the probability of being as extreme as or more extreme than 55 in the negative direction.

How Extreme? Two-Sided vs. One- Sided



This creates a problem. If you want to characterize how unusual a result is, you may want to add up these two probabilities.



If you do, you have computed a “two-sided p value.” In this case, then, we would say that an IQ of 145 has a

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“two-sided p value of .00270.”



Calculating p Values

The preceding considerations have led to some standard conventions for calculating p values (or “significance levels”) in the continuous case.

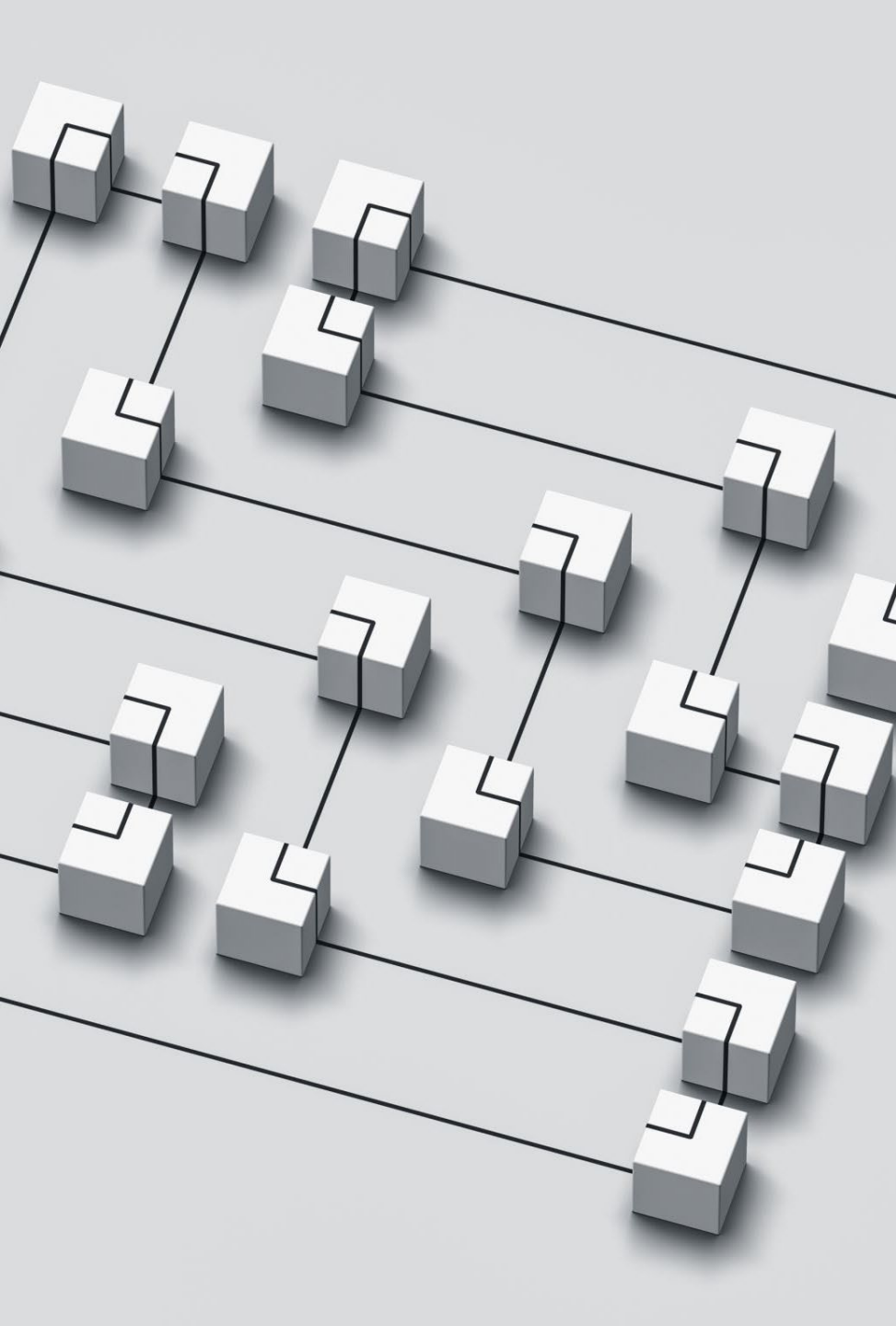
In statistical hypothesis testing, it is common to test a hypothesis about a parameter with a “test statistic” that is some function of an estimate of that parameter.

Calculating p Values

- For example, we test a hypothesis that a population mean μ is equal to 100 by examining a test statistic t that is a function of an estimate of how much μ deviates from 100.
- If the statistical null hypothesis is true, then the test statistic has a known distribution,
 - the “null sampling distribution.”

Calculating p Values

- If a value occurs that is unlikely to occur in that distribution, we consider the result as evidence against the null.
- In one-sided (or one-tailed) testing situations,
 - evidence to reject the null can come at only one end of the number line, while
- in two-sided (or two-tailed) situations, the evidence can come at either end



Calculating p Values

- Connected with this idea is a value called the significance level or p-value.
- With these ideas in mind, we can produce a rule for computing p-values
- that works well for continuous sampling distributions, but is more controversial for discrete distributions.

Calculating p Values

The p-value is the probability of getting a result as extreme or more extreme as the observed value:

- 1) If the test is one-sided,
- 2) If the test is two-sided

Calculating p Values

1) If the test is one-sided,

- the calculation is performed in the direction that
 - would result in rejection, and

the p value is the probability of

- obtaining a result more extreme than the
- observed statistic in that rejection direction.

Calculating p Values

2) If the test is two-sided, the calculation is performed toward the nearest rejection side, then the resulting probability is doubled.

The p-value is the probability of getting a result as extreme or more extreme as the observed value.

Calculating p Values

In general, the lower the p-value, the “more significant” the result.

In particular, if the p-value is less than α , we say that the result is

“significant at the α level.”

Calculating p Values

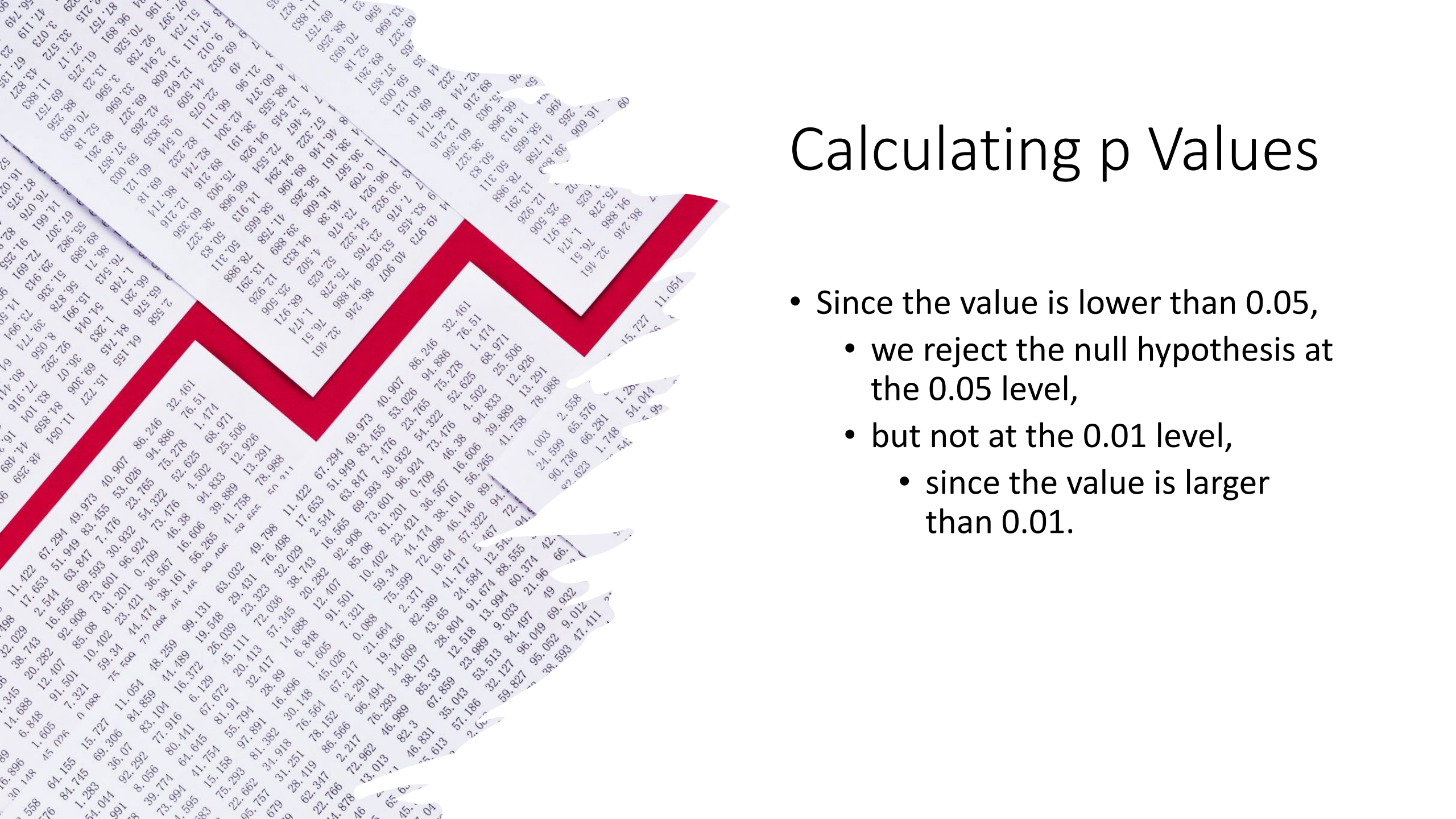
- Some people have been led to believe that this means that the lower the
 - p-value, the more powerful the effect shown in the result, but this is most
 - definitely not the case.

Calculating p Values

- To see how this works, suppose you are testing the hypothesis that
- $\mu_1 = \mu_2$ with a t-statistic with 24 degrees of freedom. The value of the statistic is 2.54.
- What is the p-value? The probability of getting a result more extreme on the positive side is
- `> 1 - pt(2.54, 24)`
- `[1] 0.008987`

Calculating p Values

- However, since this is a hypothesis about equality, it could be rejected with either a very low or a very high value of the test statistic.
- It is a two-sided test.
- So we must double the above probability to get a correct 2-sided p-value
- $> 2 * (1 - \text{pt}(2.54, 24))$
- [1] 0.01797



Calculating p Values

- Since the value is lower than 0.05,
 - we reject the null hypothesis at the 0.05 level,
 - but not at the 0.01 level,
 - since the value is larger than 0.01.



p Hacking

- Also called data dredging
 - data fishing
 - data snooping
- Occurs when researchers collect or select data or statistical analyses until nonsignificant results become significant.
 - Caused by a focus on novel, confirmatory, and statistically significant results
 - leads to substantial bias in the scientific literature
- [Illustration](#)